Subject: Programming with Java

ASSIGNMENT No : 5

TITLE : Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Outcome

Exercise

1. Create a **To-Do List application** using ArrayList to store tasks and StringBuffer to display tasks.

2. Create a **Student Course Registration System** using **ArrayList** to store the list of courses registered by a student and **StringBuffer** to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

CODE

Users > raghav_oze > Desktop > java > exp5 > todolist.java > todolist > main(String[] args)

```java
import java.util.ArrayList;
class todolist
{
    Run main | Debug main
    public static void main(String args[])
    {
        ArrayList<String> list = new ArrayList<>();
        StringBuffer task = new StringBuffer();
        task.append("complete assignments");
        list.add(task.toString());
        System.out.println("\nTo-Do List:");
        for (String item : list)
        {
            System.out.println("- " + item);
        }
        list.add("go to gym");
        list.add("sleep 7hours");
        System.out.println("\nTo-Do List:");
        for (String item : list)
        {
            System.out.println("- " + item);
        }
        list.remove(0);
        System.out.println("\nTo-Do List:");
        for (String item : list)
        {
            System.out.println("- " + item);
        }

    }
}


```

Output

```
zsh: command not found: arraylist.class
raghav_oze@raghavs-MacBook-Air-3 exp5 % java todolist.java

To-Do List:
- complete assignments

To-Do List:
- complete assignments
- go to gym
- sleep 7hours

To-Do List:
- go to gym
- sleep 7hours
raghav_oze@raghavs-MacBook-Air-3 exp5 %
```

CODE

convert.java U ● | todolist.java 1, U | courses.java 1, U ●

Users > raghav_oze > Desktop > java > exp5 > courses.java > courses > main(String[] args)

```java
/*Create a Student Course Registration System using ArrayList to store the list of courses
registered by a student and StringBuffer to generate and display the registered course list.
The application should allow users to add, remove, and view registered courses. */

import java.util.ArrayList;
class courses
{
    Run main | Debug main
    public static void main(String args[])
    {
        ArrayList<String> courseList = new ArrayList<>();
        StringBuffer registeredCourses = new StringBuffer();
        courseList.add("Mathematics");
        courseList.add("Physics");
        courseList.add("Chemistry");
        System.out.println("\nRegistered Courses:");
        for (String course : courseList)
        {
            registeredCourses.append(course).append(", ");
        }
        System.out.println(registeredCourses.toString());
        courseList.add("Biology");
        System.out.println("\nRegistered Courses");
        registeredCourses.setLength(0);
        for (String course : courseList)
        {
            registeredCourses.append(course).append(", ");
        }
        System.out.println(registeredCourses.toString());
        courseList.remove("Physics");
        System.out.println("\nRegistered Courses ");
        registeredCourses.setLength(0);
        for (String course : courseList)
        {
            registeredCourses.append(course).append(", ");
        }
        System.out.println(registeredCourses.toString());
    }
}
```

OUTPUT

```
raghav_oze@raghavs-MacBook-Air-3 exp5 % courses.java
zsh: command not found: courses.java
raghav_oze@raghavs-MacBook-Air-3 exp5 % java courses.java

Registered Courses:
Mathematics, Physics, Chemistry,

Registered Courses after adding Biology:
Mathematics, Physics, Chemistry, Biology,

Registered Courses after removing Physics:
Mathematics, Chemistry, Biology,
raghav_oze@raghavs-MacBook-Air-3 exp5 %
```

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CONCLUSION

We have successfully learnt and implemented the use of Arraylist and StringBuffer in two programs